

Kalyani Samoju

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Skills

Technologies: Machine Learning, Artificial Intelligence, Deep learning, MLops

Tools: GitHub, Grafana, MongoDB, Dagshub, HuggingFace, AWS Sagemaker

Programming Languages: Python, R, Golang

Experience

Founding AI Engineer(R&D), Vehk

06/2022 – present

With 3 years of experience worked with vehk and Led the development of machine learning models leveraging telemetry data collected from IoT devices.

- Designed and implemented data pipelines for data collection, preprocessing, and feature extraction.
- Developed predictive maintenance models to optimize asset performance and reduce downtime.
- Implemented anomaly detection algorithms to identify and mitigate security threats in real-time.
- Collaborated with cross-functional teams to integrate AI solutions into existing systems and processes.

Research Scientist Intern

02/2022 – 05/2022

- Assisted in designing and implementing secure and transparent data transactions using blockchain, ensuring the integrity of vehicle health records.
- Conducted research on emerging blockchain technologies and their potential applications in enhancing data security and transparency and running federated ML training using Blockchain

Machine Learning Intern, IIT Kharagpur

05/2021 – 08/2021

Worked on the problem, phenology information of samples from one study area using Time-Weighted Dynamic Time Warping (TWDTW) (Maus et al., 2016) to generate labels in two target areas using Sentinel-2 time-series.

- Extracted phenology information from a study area using **Time-Weighted Dynamic Time Warping (TWDTW)**.
- Utilized **Sentinel-2 time-series data** for pattern analysis.

Projects

Virtual Try on

Developed an AI-powered **Virtual Try-On system** that enables users to visualize clothing and accessories on their images in real-time. The system leverages advanced computer vision techniques, deep learning models, and high-dimensional feature extraction to ensure realistic alignment and texture preservation

OutFit Generator

Developed and deployed an **AI-powered outfit generation system**, FLUX.1-dev, which uses **LoRA (Low-Rank Adaptation)** to generate diverse and highly realistic fashion designs and outfits. The system enables users to explore creative outfit styles by leveraging advanced diffusion models for fashion trend forecasting and visualization

Education

B.Tech, International Institute of Information Technology(IIT)-SKLM ' 2023